

SignBank Enterprise — AR + AI Guidance System: Technical Proposal

1. Executive Summary

SignBank Enterprise is a cutting-edge, gesture-driven banking operations platform where users navigate and execute banking commands entirely through hand gestures via webcam. While the core gesture recognition is fully functional, adoption metrics indicate a steep learning curve for new users. Users often struggle with gesture-to-action mapping and receive no corrective feedback when mistakes occur, leading to friction in the user experience.

This proposal introduces a two-tiered enhancement system to solve this issue and drive user adoption:

1. **AR Guidance Overlay:** Real-time visual hints rendered directly on the camera feed to guide user gestures.
 2. **AI Learning Engine:** An adaptive system that tracks individual error patterns and dynamically adjusts guidance based on the user's proficiency.
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2. Problem Statement

Despite the innovative nature of gesture-driven banking, current user friction points include:

- **High Cognitive Load:** Memorizing the gesture dictionary is daunting for new users.
 - **Lack of Feedback:** When a gesture fails, the system does not explain *why* or *how to fix it*.
 - **Onboarding Drop-off:** Users who feel overwhelmed by the initial learning curve are more likely to abandon the platform.
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3. Proposed Solution

Layer 1: AR Guidance Overlay

The AR Guidance Overlay acts as a real-time, interactive tutorial built seamlessly into the user's camera feed.

- **Skeletal Tracking Overlays:** Renders semi-transparent wireframes over the user's hands to show the *ideal* hand position versus their *current* position.
- **Next-Action Indicators:** Floating tooltips or AR icons near the user's hands prompting the next required gesture (e.g., "Swipe Right to Confirm Transfer").
- **Instant Corrective Feedback:** If a gesture is partially recognized but incorrect (e.g., wrong

finger extended), the overlay highlights the specific correction in real-time.

Layer 2: AI Learning Engine

The AI Learning Engine personalizes the experience, ensuring the platform grows with the user.

- **Error Pattern Recognition:** Tracks consecutive failed gestures and identifies the specific user struggle (e.g., "User consistently struggles with the 'Transfer' gesture").
 - **Adaptive Guidance (Progressive Disclosure):**
 - **Novice Mode:** Heavy AR guidance, explicit text prompts, and slower gesture requirement thresholds.
 - **Expert Mode:** Minimal to no UI overlay. Guidance only appears if the user hesitates for more than 2 seconds.
 - **Gamified Progression:** Tracks mastery of gestures, providing visual rewards (e.g., "Transfer Gesture Mastered!") to encourage continued learning.
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4. Technical Architecture

Frontend (Client-Side)

- **Computer Vision / AR:** Utilization of WebGL and lightweight client-side ML models (e.g., MediaPipe or TensorFlow.js) to render AR overlays without network latency.
- **UI/UX Layer:** Built with React/Vue (or the existing frontend framework). Heavily leverages modern CSS (glassmorphism, micro-animations) to ensure the AR overlays feel native and premium.

Backend & AI (Server-Side)

- **Telemetry & Analytics:** A lightweight telemetry service to asynchronously log gesture success/failure rates per user session.
 - **AI Personalization Model:** A recommendation engine (using collaborative filtering or reinforcement learning) that analyzes a user's historical telemetry to adjust their "Proficiency Score" for each gesture.
 - **Database:** Extension of the existing user profile schema to store gesture proficiency metrics and AR toggle preferences.
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5. Implementation Strategy & Phases

Phase 1: Foundation & Static AR

- Implement basic AR wireframe rendering over the hand.

- Add static, contextual tooltips for the top 5 most critical banking gestures.
- **Milestone:** MVP AR Overlay deployed to staging.

Phase 2: Telemetry & AI Backend

- Deploy backend telemetry to track gesture success/failure.
- Develop the AI logic to calculate per-user proficiency scores.
- **Milestone:** Backend capable of identifying struggling users.

Phase 3: Adaptive Integration & Gamification

- Connect the AI backend to the AR frontend to enable Progressive Disclosure.
 - Implement gamified feedback micro-animations.
 - **Milestone:** Full system Beta release.
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6. Business Value & ROI

Implementing the AR + AI Guidance System directly aligns with core business objectives:

1. **Reduced Support Costs:** Fewer support tickets related to "system not working" which are often just user gesture errors.
 2. **Increased User Retention:** A gamified, frustration-free onboarding experience directly correlates to higher DAU (Daily Active Users).
 3. **Market Differentiation:** Solidifies SignBank Enterprise not just as a functional tool, but as a premium, state-of-the-art fintech experience.
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7. Millennial-Optimized UX Design Principles

[!IMPORTANT] The target demographic expects a flawless, modern aesthetic. The AR overlay must not feel like a clunky tutorial.

- **Micro-animations:** Smooth, 60fps transitions when AR elements appear or disappear.
- **Glassmorphism UI:** AR tooltips will use frosted glass effects to remain legible without completely obscuring the camera feed.
- **Positive Reinforcement:** Celebrate successes immediately with subtle, rewarding visual cues, mimicking modern video game UI mechanics.
- **Zero-Friction Opt-out:** Users can instantly dismiss guidance with a simple, universal gesture (e.g., "Swiping away" the tutorial).